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Energy, greenhouse gas, and water life cycle analysis of lithium carbonate and lithium hydroxide monohydrate from brine and ore resources and their use in lithium ion battery cathodes and lithium ion batteries

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ABSTRACT

Life cycle analyses (LCAs) were conducted for battery-grade lithium carbonate (Li_2CO_3) and lithium hydroxide monohydrate ($LiOH•H_2O$) produced from Chilean brines (Salar de Atacama) and Australian spodumene ores. The LCA was also extended beyond the production of Li_2CO_3 and $LiOH•H_2O$ to include battery cathode materials as well as full automotive traction batteries to observe the effect that the lithium production pathways had on these end products. The LCA here covers material, water, and energy flows associated with lithium acquisition; lithium concentration; production of lithium chemicals, battery cathode powders, and batteries; and associated transportation activities along the supply chain. Based on battery cathode material, the difference in lithium source represents a difference of up to 20% for NMC811 cathode greenhouse gases (GHGs) and up to 45% for NMC622 cathode GHGs. For full batteries, this represents a difference of up to 9% for NMC811 batteries and 20% for NMC622 batteries. Production of Li_2CO_3 from brine-based resources had less life cycle GHG emissions and freshwater consumption per tonne of Li_2CO_3 than Li_2CO_3 from ore-based resources. And $LiOH•H_2O$ produced from brine-based lithium also had less life cycle GHG emissions and freshwater consumption per tonne of $LiOH•H_2O$ than $LiOH•H_2O$ from ore-based resources.

1. Introduction

Lithium is an essential element for the rechargeable battery market. The U.S. Geological Survey (USGS) estimates that batteries constitute 65% of the end-use market for lithium (USGS 2020). These batteries are a driving force in the modern economy, from powering personal electronics to grid storage systems and automobiles. Lithium is not a rare resource; it is abundant in the lithosphere and available from brines and hard rock deposits (Jaskula, 2018). It is, however, considered a critical material by the U.S. Department of Interior and the European Commission (Miatto et al., 2020, U.S. Department of the Interior, 2018a, European Commission, 2020). Ambrose and Kendall (Ambrose and Kendall, 2020b) describe several challenges to the lithium market and lithium production. They highlight that while lithium has a high rate of end use in electrified vehicles to improve transportation sustainability, yet sustainability challenges remain for lithium extraction as for any natural resource (Ambrose and Kendall, 2020a).

According to the USGS, lithium has historically been acquired from either continental brines or hard-rock minerals. Chile has long been a leading producer of lithium carbonate (Li_2CO_3), with production from two Salar de Atacama (Atacama Salt Flat) brine operations next to the Andes Mountains. Lithium concentrates are transported for processing to two Li_2CO_3 plants and one lithium hydroxide monohydrate ($LiOH•H_2O$) plant (Jaskula, 2018) in Chile. Argentina also produces Li_2CO_3 and lithium chloride from brines in the Andes Mountains. The USGS notes that Australia is the leading producer of lithium mineral concentrates. Australian lithium mining has grown significantly in the past decade and now accounts for 54% of total world lithium production, but it has historically shipped its concentrates to other locations: All refining takes place in China, while exports to other countries are for

Abbreviations: CO_2e, Carbon dioxide equivalent; GHG, Greenhouse gas; LCA, Life cycle analysis; LCE, Lithium carbonate equivalent; LCI, Life cycle inventory; Li_2CO_3, Lithium carbonate; $LiOH•H_2O$, Lithium hydroxide monohydrate; NMC622, $LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$; NMC811, $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$; PM10, Particulate matter 10 micrometers or less in diameter; PM2.5, Particulate matter 2.5 micrometers or less in diameter; T&D, Transmission and distribution; USGS, United States Geological Survey.

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direct use in ceramics, glass and other industrial uses (USGS 2019).

According to the USGS, total worldwide lithium production in 2019 was 77,000 tonnes lithium, or 410,000 tonnes lithium carbonate equivalent (LCE) (USGS 2020). The year-over-year trends have shown consistent increases in global supply of lithium between 2006 and 2016 (Jaskula, 2018). Jaskula also reports that worldwide lithium consumption between 2006 and 2016 grew from 15,000 tonnes Li (80,000 tonnes LCE) to more than 35,000 tonnes Li (185,000 tonnes LCE). Bloomberg New Energy Finance projects that production of lithium in 2030 will be 1.5 million tonnes LCE (~280,000 tonnes lithium), based on nameplate capacity and de-risked supply (Lu and Frith, 2021), and projects the consumption of lithium to range between 1.3 and 2.0 million tonnes LCE (240,000−375,000 tonnes Li). One of the main drivers of this projected increase in lithium demand is its use in electric vehicles.

Argonne National Laboratory worked with Sociedad Quimica y Minera S.A. (SQM), a global leader in lithium production, to conduct a process-based life cycle analysis (LCA) for Li_2CO_3 and $LiOH•H_2O$. A life cycle analysis uses detailed life cycle inventory (LCI) data: a collection of the detailed input and output flows for a product along its supply chain, including raw materials, energy, and water, and process emissions. An LCA uses the LCI data to evaluate the environmental effects that the product has throughout its supply chain and all production stages: resource extraction and processing, manufacturing, distribution, use, recycling, and disposal.

The analysis here is for the production of Li_2CO_3 and $LiOH•H_2O$. The lithium producer shared data regarding production from the Salar de Atacama, where subsurface lithium brine is pumped to drying ponds for concentration and then transported to its plants for production of Li_2CO_3 and $LiOH•H_2O$. The LCI data for ore-based operations are based on informed assumptions for production in western Australia, in which spodumene ore would be extracted, locally concentrated, and then transported to coastal western Australia for worldwide shipping. We augment that data with data from an environmental impact assessment report for a cathode plant in China by Tianqi Lithium, a major cathode producer, for production of Li_2CO_3 and $LiOH•H_2O$ from spodumene ore (Tianqi Lithium Corporation, 2017).

This LCA uses Argonne's open-source modeling tool, GREET® (Greenhouse gases Regulated Emissions and Energy in Technologies). GREET models the life cycle energy consumption, water consumption, and emission of greenhouse gases (GHGs) and air pollutants associated with the production of materials. Inputs for a GREET LCA include all upstream energy, water, and emissions for the supply chains of not only the finished product but all energy sources and materials used in its production and transportation. Note that we utilize GREET's framework for calculation and leverage its background data for energy and material burdens associated with some resources. However, throughout this article, data for location specific activities are used. For example, the electrical grids of Chile, Australia, and China are integrated to be location specific along with several other regional specific details highlighted in the text and the Support Information.

Water consumption is an important consideration in environmental analysis. This study follows the conventions of supply chain LCA (Wang et al., 2019, Lampert et al., 2016), which evaluates consumptive use of freshwater water resources. Consumptive water refers to fresh water that is brought into an industrial system and not returned to the freshwater system because it is either evaporated to the atmosphere or contained within the final product. In general, freshwater consumption may involve both blue water and green water (e.g., water consumption from rainfall and soil moisture sourced from rainwater). This study only considers blue water, that is, fresh water mainly from surface and ground water, the two sources that are subject to water resource depletion by human activities. On the other hand, the part of the green water from rainfall is usually re-occurring and not subject to resource depletion. However, non-fresh water, such as brine, cannot be used directly for human activities and is differentiated from fresh water consumption in water resource analysis.

Recently, Schomberg et al. (2021) conducted a spatially explicit LCA to determine a water scarcity footprint based on the AWARE scarcity system for lithium ion batteries and included evaporated water originating from brine in that footprint. That approach is different from the one used here. In our analysis, we include fresh water pumped from groundwater east of the Salars to compensate for brine evaporation during lithium recovery, while the evaporated water in Schomberg et al. includes water molecules from both ground water and brine. Brine, the type of water from which lithium can be derived, has much higher salinity than fresh water. Based on producer-supplied data, Salar de Atacama brine has ~350 g/L of total dissolved solids, which is ~10 times more than seawater. This type of water is not suitable for either agricultural or domestic use (Ayers and Westcot, 1985, World Health Organization 2008), so we did not include the water molecules originating from brine in our fresh water consumption. Nonetheless, both brine and groundwater are key factors of the ecosystem in the Salar de Atacama. The hydrogeological dynamics of both water systems in the context of lithium production were studied in detail by Marazuela et al. (2018a,2018b) and Aravena et al. (2019).

The GREET model has an extensive LCA capability for batteries. The model contains inventory data for the materials and energy used in battery production, including lithium, nickel, cobalt, manganese, and aluminum, in the various chemical forms relevant to battery manufacturing. It also contains data for many other battery and battery cathode-relevant materials, such as lithium hexafluorophosphate, ethylene carbonate, dimethyl carbonate, and N-methyl-2-pyrrolidone. GREET considers the materials and energy required for all stages of battery production, from the extraction of primary materials through

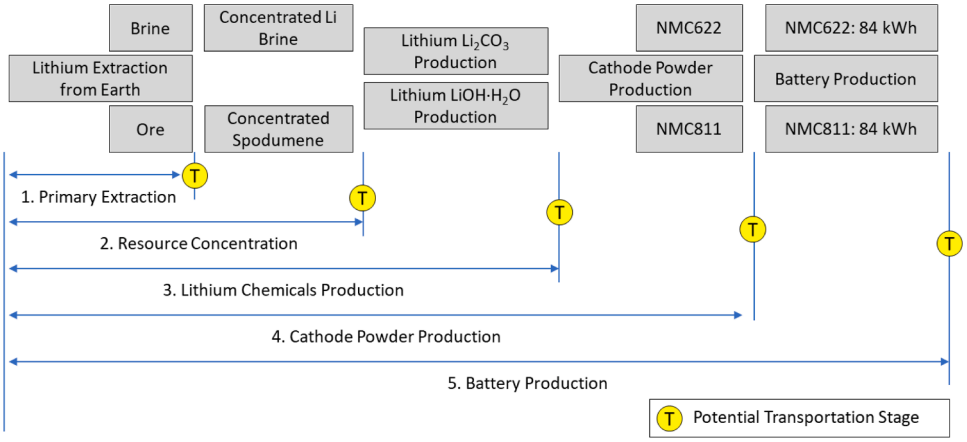


Fig. 1. The stages of LCA evaluation in this study.

battery cell and battery pack production to the existence of vehicle-size batteries. These GREET features were used in this study to extend lithium LCA results from the detailed analysis of lithium supply chain here into battery LCAs.

This study reports a detailed LCA for the production of Li_2CO_3 and $LiOH•H_2O$ from brine, based on primary data, and for the production of Li_2CO_3 and $LiOH•H_2O$ from ore, based on a combination of the literature, engineering estimation, and primary data. The scope of this LCA is from the mine or well to the chemical product post-delivery gate (Li_2CO_3 or $LiOH•H_2O$) for two different pathways based on resource origin (brine or ore). The functional unit is 1 tonne of battery-grade product (Li_2CO_3 or $LiOH•H_2O$). We also conducted LCAs for battery cathode material and battery production from the pathways identified in order to put the lithium LCA results into the broad perspective of electric vehicle batteries and electric vehicles. We assume that the lithium-ion batteries based on $LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$ (NMC622) and $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811) that are produced are those used in electric vehicles with battery driving ranges of 300 miles (482 km), because these are becoming common battery sizes in the automotive market. These batteries require the production of the NMC622 and NMC811 cathode materials. We assume that NMC622 is produced using Li_2CO_3, and NMC811 is produced using $LiOH•H_2O$. GREET contains all necessary data for an LCA of both cathode material and batteries. The increasing prevalence of electric vehicles coupled with their use of increasingly long-range batteries (greater kWh) will continue to increase the importance of properly characterizing the environmental burdens associated with extracting lithium since it is an important component of those batteries, and batteries will represent an increasingly large portion of a vehicle's life cycle burdens (especially as the electrical grid decarbonizes).

In this analysis, we evaluate energy, GHG emissions, and water consumption in the life cycle of lithium from its extraction from the earth through its integration into a vehicle battery. Fig. 1 provides a simplified illustration of the six stages of analysis. Stages 1 and 2 represent the extraction of the lithium resource from the earth, in either brine or spodumene ore form, and the concentration of the lithium to a level at which it can be used for additional chemical transformation. Stage 3 includes lithium chemicals production, whether that be Li_2CO_3 or $LiOH•H_2O$ (in some pathways, $LiOH•H_2O$ is produced from Li_2CO_3, not directly from the concentrated brine or spodumene ore). Stage 4 extends the analysis to cathode powder production, whether that be NMC622 or NMC811, which we assume here to be derived from Li_2CO_3 and $LiOH•H_2O$, respectively. Stage 5 is battery production. This diagram helps position each analysis stage for the reader and will be referred to throughout the article. More detail regarding the boundaries of each stage, material and energy inputs, and product outputs are provided in the main text as well as the Supplemental Information.

2. Lithium extraction from brine

Lithium has historically been produced from subterranean brines in the Andean mountain region of Chile in the area known as the "Lithium Triangle," which also includes parts of Bolivia and Argentina. There are also several other lithium-rich brine locations throughout the world, and each brine has a unique chemical composition, which necessitates different processes for isolating the desired end products (Wietelmann and Steinbild, 2014). This LCA considers the brines in the Atacama Desert of Chile, known as the Salar de Atacama. The brines are concentrated, processed into battery grade Li_2CO_3 and $LiOH•H_2O$, and shipped worldwide for processing into battery cathode materials and, eventually, batteries.

2.1. Concentrated lithium brine production from Salar de Atacama

The Salar de Atacama is a small, unique region located east of the Atacama Desert at 2300 m above sea level. It is surrounded by mountainous regions, with the Andes mountain range to the east and the Domeyko range to the west. While the Atacama Desert is hyper-arid, the Salar de Atacama is classified as a marginal high desert climate (Koppen, 1936). The Salar de Atacama basin receives between 5 and 120 mm of annual rainfall depending on the specific location. To produce concentrated brine, a low-concentration brine (~0.17% lithium concentration, which is 70% water by mass and the remainder other minerals) is pumped to the surface and into a drying pond where it can be exposed to the sun and heat to facilitate evaporation (Stage 1 in Fig. 1). These ponds are in the southern part of the Salar de Atacama, which receives less rain, thereby facilitating effective evaporation.

At the Salar de Atacama operation, there are two separate pathways: lithium and potash separation (Figure S.1, Supplemental Information [SI]). The lithium pathway has two flows of potassium into the potash pathway, while the potash pathway has no flows back to the lithium pathway. Because potash is a coproduct of the overall system, we need to allocate the energy consumed in the production of potash into our modeling. While we largely focus on the specifics of the lithium production pathway, we also track the flow of potassium chloride (KCl) to the potash pathways. We also need to allocate energy and water associated with the various site operations not specific to either lithium or potash production that were detailed by the lithium producer.

After the brine is pumped to the surface, it flows through several ponds before final dispatch as lithium chloride brine with a lithium concentration of ~6% (Stage 2 in Fig. 1). The extracted brine is managed in the pond system for approximately one year while water is successively evaporated and lithium is concentrated. In the process, various salts are extracted from the ponds, which facilitates the removal of impurities from the brine and further increases the lithium concentration. The final brine is shipped for conversion to Li_2CO_3.

We tracked the flow of LiCl brine and KCl, the value products for this system. Proprietary information from the producer specifies the quantity of KCl associated with the flows. In addition to the flow of KCl as a valuable coproduct, there are losses of lithium, along with water and other minerals in the system, to system infiltration (i.e., the flow of brine back into the earth). Lithium is also stored within the system in the form of brine, harvested salt, and impregnated salts. We obtained data for the input and output flows for 2018 for each pond, allowing process-level analysis of this LCA effort.

Allocation of energy use and emission burdens among different products from one system is an important issue affecting product-specific LCA results. The allocation can be done at the process level with detailed input and output data and associated energy use and emissions. In addition to a process-level approach, we examined two other approaches to evaluating the energy and water consumption used to concentrate lithium brine: the facility level and the product level. The facility-level approach is the type of analysis typical of LCIs conducted with only high-level publicly available data. Illustrated in Figure S.2 (see SI), the facility-level approach aggregates energy, material, and water entering the Salar de Atacama system and allocates them across the outputs from the facility—concentrated lithium brine and potash—on a mass or economic-value basis. (See Appendix A in the SI for an explanation of economic value allocation.) In this analysis we include both the mass- and value-allocation approaches for the facility-level method.

In the product-level method shown in Figure S.3, the energy and water for primary brine pumping are allocated on a mass basis between the lithium and potash production lines. The lines are treated as separate entities in which only the energy, material, and water inputs for those lines are attributed to their associated product. One exception is nonspecific site operations energy and water consumption, which are allocated on a final output mass basis to each system.

Different assumptions in allocation can cause significant differences in LCA findings. Here, we have access to detailed data allowing us to utilize the process-, product- and facility-level (mass, and value) allocation approaches to illustrate how these allocation choices may affect results of not only the main product (concentrated brine), but also the

Table 1
Material and energy flow summary by allocation methods for per tonne of concentrated brine (6% lithium) produced in the Salar de Atacama.

Per tonne concentrated brine	Facility Level (mass)	Facility Level (value)	Product Level	Process Level	Unit
Materials input					
Lithium brine	24.1	24.1	24.1	24.1	tonne
Fresh water	2.40	5.94	3.72	2.35	m^3
Energy input					
Electricity	307	760	624	472	MJ
Diesel	265	657	346	327	MJ
Coproduct					
Potash	8.52	8.52	8.52	8.52	tonne
By-product (as stock in the system)					
Lithium	0.80	0.80	0.80	0.80	tonne as LCE

Table 2
Material and energy flow summary per tonne of Li_2CO_3.

Per tonne Li_2CO_3	Input	Unit
Materials input		
Lithium brine (6% Li)	4.00	tonne
Soda ash	2.0	tonne
Other[a]	0.08	tonne
Energy input		
Electricity	1500	MJ
Diesel	400	MJ
Natural gas	2800	MJ
Noncombustion emissions		
PM 10	700	g
PM 2.5	400	g

[a] Includes H_2SO_4, HCl, lime, solvent, and alcohol.

final products (lithium chemicals, cathodes, and batteries). Throughout this analysis we present results of the ore-based lithium considering each of the four noted allocation approaches. It is worth noting that we are not describing physical process differences, rather differences in the known information about the production processes.

Material and energy requirements for each of the evaluation approaches are summarized in Table 1. We assume the stored lithium is treated as stock rather than a coproduct and may eventually be recovered for value. Following Dunn et al. (2014), we assume the diesel fuel used on site is split between off-road vehicles that transport materials and manage the site (23%) and reciprocating engines that provide power for operations (77%). The electricity used for this analysis is the average Chilean electricity grid (International Energy Agency and Data, 2020). We used market prices to determine that concentrated lithium brine has three times the value of potash.

2.2. Li_2CO_3 production from concentrated lithium brine

The concentrated lithium brine is trucked 232 km from Salar de Atacama to Antofagasta, Chile, for refinement and processing (Stage 3 of Fig. 1). Each truck carries 24.5 tonnes of payload. At the Li_2CO_3 production plant, boron is removed from the brine, which then undergoes purification and carbonation (Wietelmann and Steinbild, 2014, Dunn et al., 2014). The boron extraction facility consumes hydrogen chloride (HCl), alcohol, sulfuric acid (H_2SO_4), and an organic solvent to generate a boron solution. The treated lithium brine then proceeds to extraction phases that remove magnesium by adding soda ash (Na_2CO_3) and lime (CaO). The purified brine is combined with soda ash for a precipitation reaction that yields Li_2CO_3 as a solid. The resulting product is washed, filtered, and dried prior to compaction and packaging. The process data provided by the producer confirms this operation, which is represented in a simplified flowchart in Figure S.4.

Table 3
Material and energy flow summary per tonne of $LiOH{\bullet}H_2O$.

Per tonne $LiOH{\bullet}H_2O$	Input	Unit
Materials input		
Li_2CO_3	1.05	tonne
CaO	1.15	tonne
Water consumed	0.5	m^3
Energy input		
Electricity	5000	MJ
Diesel	3000	MJ
Natural gas	21,000	MJ
Noncombustion emissions		
PM 10	100	g
PM 2.5	50	g

Using producer data, we determined the total material, energy, and water consumed to produce Li_2CO_3. The entire Li_2CO_3 production process is treated at the facility level since the facility produces a single product—Li_2CO_3—and process-level analysis is not needed. In addition to material, water, and energy inputs, we were provided particulate matter emissions data for the facility's operation. Material and energy flows for Li_2CO_3 production from lithium brine are summarized in Table 2. Water, while used in the process, is a fully recycled stream and not consumed and therefore not included in the LCI.

Since only Li_2CO_3 is produced in this scenario, there is no need to allocate to products. However, a portion of the Li_2CO_3 becomes an input for the production of $LiOH{\bullet}H_2O$, which we describe in the next section.

2.3. $LiOH{\bullet}H_2O$ production from brine-based Li_2CO_3

Li_2CO_3 is reacted in a series of stages with a mixture of CaO and water to produce $LiOH{\bullet}H_2O$ (Stage 3 in Fig. 1). Again, since $LiOH{\bullet}H_2O$ is the only product from the facility, process-level analysis is not needed; we analyzed data at the facility level and were able to determine the requisite energy and material inputs for $LiOH{\bullet}H_2O$ production. Note that for the provided data there are two facilities at the same plant for Li_2CO_3 and $LiOH{\bullet}H_2O$ production, respectively, but the facilities are separated, and the process data provided for each is unique.

Figure S.5 shows a flowchart for this aggregate production process. Using producer data, we determined the total facility-level material, energy, and water demand to produce $LiOH{\bullet}H_2O$. Emissions data provided were used to calculate non-combustion process emissions. The material and energy flows for $LiOH{\bullet}H_2O$ production from lithium brine are summarized in Table 3. Water is consumed in this process. Again, since only $LiOH{\bullet}H_2O$ is produced, there is no need to allocate to products.

2.4. Electricity generation mix, transportation, and water consumption for lithium brine and its products

The Chilean electrical grid was historically two separate grids: one for northern Chile and another for the rest of the nation. In 2018, the two grids were linked via two high-voltage transmission lines. Production of lithium concentrate occurs in the remote Salar de Atacama, and Li_2CO_3 and $LiOH{\bullet}H_2O$ are produced near the port of Antofagasta. Both locations are in the northern region of Chile, so they are more likely to use electricity that is produced inside, rather than outside, their region. However, current reporting of electricity by the Chilean Ministry of Energy does not provide enough granularity to ascertain the grid composition of the northern region (Ministerio de Energía 2018). Thus it is reasonable to use the generalized Chilean electrical grid composition, given in Table S.1, to calculate upstream environmental impacts of electricity used when making lithium products (International Energy Agency and Data, 2020). Transmission and distribution loss factor for Chile (6.7%) is reported in Table S.1 (The World Bank, 2020). GREET's default

production pathways and fuel combustion emission factors were assumed for diesel and natural gas consumption, since the production and combustion technologies of diesel and natural gas are unlikely to exhibit strong regional variation (Wang et al., 2019).

Using provided water data, we examined water consumption at the facility (mass and economic value allocations) at the product and process levels and provide the results throughout this report. In the Results section we give the water consumption results for each of the output products. Additional water data for brine operations are provided in SI Section 2.4.

Li_2CO_3 and $LiOH•H_2O$ production requires auxiliary chemicals during the refining process. These include soda ash, H_2SO_4, HCl, alcohol, organic solvent, and CaO, all of which are modeled in GREET. The default production and transportation amounts for these resources are used in this analysis with an assumed Chilean production basis. The exception is CaO, which we assume is produced in the United States and transported to Chile via ship and truck, with an assumed distance of 5000 and 200 miles (8047 and 322 km), respectively. This CaO proxy was utilized for anonymization purposes.

In Chile, concentrated lithium brine is transported to the chemicals production site via truck (232 km, 24.5 tonne payload), then additional transportation is required to move Li_2CO_3 and $LiOH•H_2O$ to port for worldwide shipment (40 km, 20 tonne payload). We determined a shipment-weighted average distance for worldwide shipping using producer data. Transportation needs are summarized in Table S.2.

3. Lithium extraction from ore

Lithium is abundant in several different ore structures throughout the world. Spodumene ore is abundant in western Australia and has been a longtime global source of lithium (Jaskula, 2018, Ambrose and Kendall, 2020b, Wietelmann and Steinbild, 2014). Spodumene ($LiAlSi_2O_6$) contains up to 8% Li_2O (by mass) and can be refined to obtain lithium chemicals such as Li_2CO_3 or $LiOH•H_2O$. We examined the production of concentrated spodumene from lithium-containing ores in western Australia and its conversion into Li_2CO_3 or $LiOH•H_2O$ in China. More details on the input energy and transportation modeling can be found in Section 3 of the Supplemental Information.

Generally, lithium ore is extracted from the earth and reduced in size through a series of crushing and milling stages (Figure S.6). A series of flotation steps follows to separate the valuable spodumene from various tailings (Wietelmann and Steinbild, 2014). The spodumene concentrate is then transported to its next stage of production. The spodumene concentrate is calcinated and then leached with sulfuric acid to produce lithium sulfate, from which Li_2CO_3 and/or $LiOH•H_2O$ is produced. The production processes are treated at the facility level based on the available aggregate data. In this section we report the energy, water, and material inputs for the final production of Li_2CO_3 and $LiOH•H_2O$ from spodumene ore.

3.1. Spodumene concentrate production from Western Australian spodumene ore

We developed an LCI data set for lithium ore mining and processing parameters in western Australia. Site assumptions included the need to mine 14 million tonnes of material annually with an annual yield of 2 million tonnes of ore, which we assumed contained ~0.8%–0.9% Li_2O. The primary operations of the mining site would be excavating and moving materials with diesel trucks (Stage 1 in Fig. 1). We estimated diesel consumption by these trucks on the basis of continuous operation of 10 trucks. An estimated 1 million m^3 of water would be consumed annually, some of which may be discharged to a water system.

We assumed that the mined spodumene would be concentrated (Stage 2 in Fig. 1) in a facility designed to have an independent electricity plant operating on diesel fuel. The plant would receive the ore and concentrate it through a series of size reduction, separation, and purification operations, which would consume reagents in relatively low quantities as well as water. The facility will produce concentrated spodumene ore. We assume that the concentrated spodumene has a minimum of 63% pure spodumene with a remaining mass of quartz and feldspar. This yields a minimum ~5% concentration of Li_2O by mass for the delivered spodumene ore. Table 4 presents our assumed material and energy inputs to mine and produce concentrated spodumene from western Australia mines.

Table 4
Material, energy, and water inputs per tonne concentrated spodumene produced.

Per tonne concentrated spodumene	Quantity	Units
Materials input		
Spodumene ore (0.8–0.9% conc.)	4.5	tonne
Other[a]	0.015	tonne
Fresh water	3	m^3
Energy input		
Diesel	4500	MJ

[a] Includes sodium carbonate and a dispersant.

Table 5
Material and energy flow summary per tonne of LiOH•H2O and Li2CO3 produced in China from Australian spodumene concentrate.

	Input Per tonne $LiOH•H_2O$	Input Per tonne Li_2CO_3	Unit
Materials input			
Spodumene concentrate (6% Li_2O)	6.42	7.3	tonne
H_2SO_4 (98% conc.)	1.52	1.71	tonne
Na_2CO_3 (98.8% conc.)	0.025	2.05	tonne
NaOH (96% conc.)	1.18	0.05	tonne
$CaCO_3$ (≥ 98% conc.)	0.6	0.7	tonne
Fresh water	11.24	40	m^3
Energy input			
Electricity (China grid)	12,600	6480	MJ
Coal (for $LiOH•H_2O$)[a]	71,343	–	MJ
Coal (for Li_2CO_3)[b]	–	135,890	MJ
By-product output			
Na_2SO_4	1.72	1.92	tonne

[a] 54% for steam and 46% for kiln.
[b] 39% for steam and 61% for kiln.

3.2. Li_2CO_3 and $LiOH•H_2O$ production from spodumene concentrate (China)

Once spodumene has been concentrated, lithium compounds (LiOH•H2O or Li_2CO_3) can be produced from it (Stage 3 in Fig. 1). A cathode plant environmental impact assessment report produced by the Tianqi Lithium Corporation, a major cathode producer in China, details the inputs of material, energy, and water it uses to produce 1 tonne of $LiOH•H_2O$ or Li_2CO_3 [10]. Their report assumes concentrated spodumene ore from western Australia at 6% concentration. Since we assumed the concentrated spodumene ore has a minimum of 5% concentration, it is sufficient for the production process here. We assume that the spodumene is shipped from coastal western Australia to China and transported 100 miles (161 km) by truck (33 tonne payload) to a lithium compounds facility. Note that these shipping estimations are not specific to the reporting company but to concentrated production regions in China.

The scope of the Tianqi-reported data allows for a facility-level analysis of the material, water, and energy needed to produce 1 tonne of $LiOH•H_2O$ or Li_2CO_3 from spodumene concentrate (Table 5). A significant amount of coal is used for process heat, split between use as fuel for an acid-roasting kiln and for steam production. Tianqi reports that it uses 12 tonnes of steam per tonne of $LiOH•H_2O$ produced and 16.5 tonnes of steam per tonne of Li_2CO_3 produced (Tianqi Lithium Corporation, 2017). We convert this to coal energy in Appendix B of the SI. We

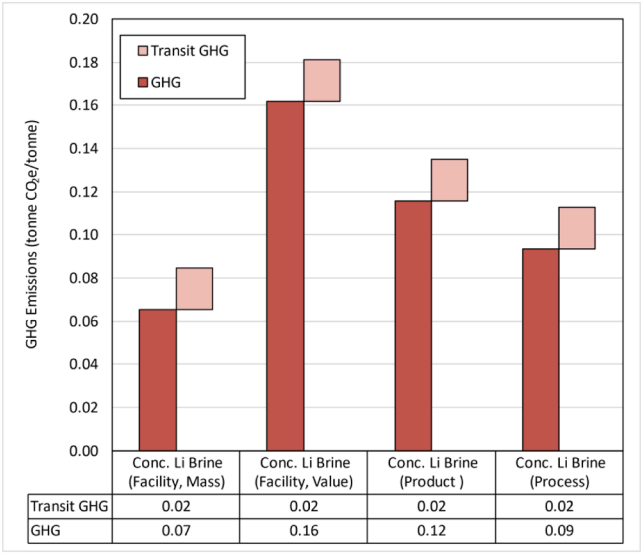


Fig. 2. Cradle-to-gate life cycle GHG results for concentrated brine production based on four allocation approaches.

conducted a sensitivity analysis to examine the life cycle results of these processes if natural gas were used instead of coal in Section 4.7 of the SI. Note also that Na_2SO_4 is produced as a by-product of this process. We determined the quantity of the by-product but did not allocate any burdens to it. We discuss the rationale in Section 3.2 of the SI.

The electricity mixes assumed for Australia and China are provided in Table S.3 along with details of spodumene concentrate transportation (see Section 3.6, Table S.4, and Table S.5 of the SI). Default GREET material production assumptions were used here. These are global commodities and the production processes for these materials are well established.

4. Results and discussion

The following sections detail the LCI and LCA results for producing lithium-ion batteries and their various precursor pathways. We begin by presenting results for the cradle-to-gate production of concentrated lithium brine considering four allocation approaches (Section 4.1) and concentrate spodumene (Section 4.2). Section 4.3 presents LCA results for producing Li_2CO_3 from either brine- or the ore-based Li, while Section 4.4 shows the production of $LiOH{\bullet}H_2O$ from brine- or the ore-based Li. Section 4.5 presents the results of cathode production from the different lithium sources (comparing the different allocation methods), and finally, Section 4.6 shows the results of the battery production considering the different Li origination type and allocation method. The analysis within each section is cradle-to-gate, thus results are inclusive of all upstream life-cycle stages to that point. The outputs of each stage serve as the appropriate input to the following stages.

4.1. Cradle-to-gate LCA of concentrated lithium brine

We conducted a cradle-to-delivery-gate LCA for producing 1 tonne of concentrated lithium brine from Salar de Atacama brine based on implementing Table 1 data in the GREET model. We determined the total energy, material, and water consumed along with air pollutants produced from both direct inputs and upstream processes (i.e., fuel resource extraction, refining, auxiliary materials production). Air pollutants considered in GREET included CO_2, CH_4, N_2O, VOC, CO, NO_x, SO_x, PM_{10}, $PM_{2.5}$, black carbon, and organic carbon. Fig. 2 presents the results for life cycle GHG emissions while Figure S.8 and Figure S.9 show the results for life cycle energy use and life cycle water consumption, respectively.

Fig. 2 shows the effect of the four different allocation approaches on the final results. The facility-level value allocation indicates the largest GHG burden, this highlights that the value difference between concentrated lithium brine and potash would (at today's market prices) allocate more of the output burden to the concentrated lithium brine, whereas the facility-level mass allocation indicates a lower associated GHG burden as it allocates all emissions equally across the mass outputs of the two co-products. The product-based allocation approach allocates a greater GHG burden to concentrated lithium brine than the facility-level mass approach since the facility-level mass approach apportions a significant amount of the initial pumping energy inputs to potash (Figure S.3). Finally, the process-based methodology leverages the most detailed information for the identified approaches as it allocates each process stage input by the stage's outputs. We see from these results that the various process stages (mostly pumping brine from one pond to another) are the primary drivers of energy consumption, GHG emissions, and water consumption. Transportation represents a meaningful contribution, on the order of 10%, for energy and GHG burdens, but is minimal

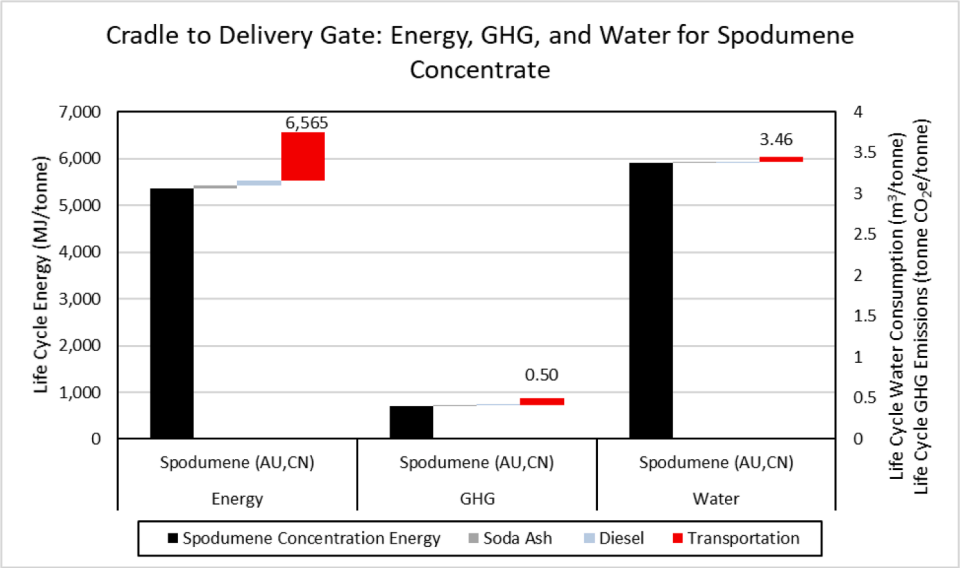


Fig. 3. Cradle-to-gate life cycle energy, GHG, and water results for concentrated spodumene production based on process and material inputs.

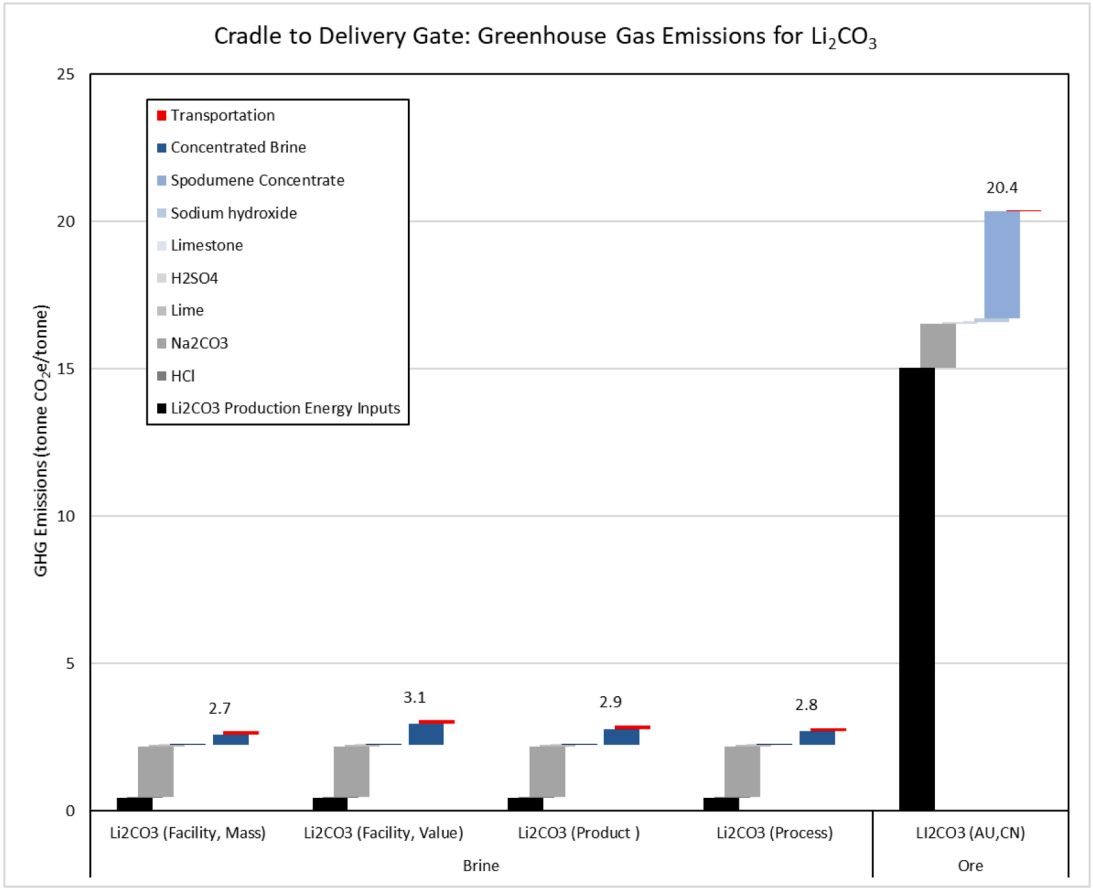


Fig. 4. Cradle-to-gate life cycle GHG results by input and process for Li_2CO_3 production based on Chilean brine and Australian spodumene ore.

for water consumption. It is important to recall that the results shown are for potential choices made with respect to allocation approach, which often depends upon data access. We present data in this way to illustrate this critical element of LCA and recommend process-level allocation to the extent possible by available data.

4.2. Cradle-to-gate LCA of concentrated spodumene

The LCI for the production of spodumene concentrate is shown in Table 4. We conducted a cradle-to-gate LCA based on these data using the GREET model. The results for total energy, GHG, and water are presented in Fig. 3. It shows the energy, GHG, and water burden categories together in a waterfall chart showing the effects of spodumene concentration energy inputs, material inputs, and transportation of the spodumene concentrate. We see that the energy inputs into the mining and concentration stages (in black) are the main drivers of embodied impacts (>80%), while transportation can also affect outcomes, particularly GHG emissions.

4.3. Cradle-to-gate LCA of Li_2CO_3

The processes of producing of Li_2CO_3 from lithium extracted from brine and ore are detailed in Sections 2.2 and 3.2, respectively. The energy, materials, and water requirements are given in Tables 2, 4, and 5 for brine and ore, respectively. We prepared the results of the cradle-to-delivery-gate LCA results for 1 tonne of Li_2CO_3 produced from brine originating in the Salar de Atacama and processed in Chile (Table S.6) and for Li_2CO_3 produced from spodumene ore extracted in western Australia (AU) and processed in China (CN), as shown in Table S.7.

Figure S.10 and Figure S.11 show the cradle-to-delivery-gate LCA results for energy use in the Chilean brine pathway, with four allocation approaches, and the Australia–China spodumene ore pathway. We see that the carbonate production stage in China is very energy intensive and performs worse than the four Chilean allocation approaches, and that the spodumene concentration stage is highly impactful, as is the brine concentration stage. We also see that the input of Na_2CO_3 has a meaningful impact on energy use. Note that the transit stage is only for the transit of analyzed product (Li_2CO_3). Fig. 4 shows the GHG emissions, which track closely with the energy consumption results, while Figure S.12 presents the results for water, where again the ore-based pathway performs worse than the brine-based pathway. In Fig. 4 we see that for the brine-based pathways, Na_2CO_3 is the primary driver of GHG emissions followed by the energy to produce Li_2CO_3 and the concentrated brine (depending upon brine allocation approach). But for the ore-based lithium, Li_2CO_3 production is the main driver followed by the spodumene concentrate. Production of Li_2CO_3 in China is modeled on a facility that uses coal for heat, transitioning that to natural gas would likely reduce GHG emissions (see SI Section 4.7.1).

The GHG emissions findings for brine-based production of Li_2CO_3 are consistent with the literature, while the findings for the production route from ore differ. Previous work by Dunn et al., formulated into the GREET model, suggested GHG emissions of 3.8 tonnes of CO_2e per tonne of Li_2CO_3 from brine, while Ambrose and Kendall found that Li_2CO_3 from brine resulted in 3.06 tonnes of CO_2e per tonne of Li_2CO_3 [7,11,20].

The production of Li_2CO_3 from brine differs from its production from ore. Ambrose and Kendall found that Li_2CO_3 from ore creates 2.28 tonnes of CO_2e per tonne of Li_2CO_3 [7]. The present study reflects new information, finding distinct increases in CO_2e per tonne of Li_2CO_3 from spodumene production, in use of Na_2CO_3, and in the energy inputs for the Li_2CO_3 production itself.

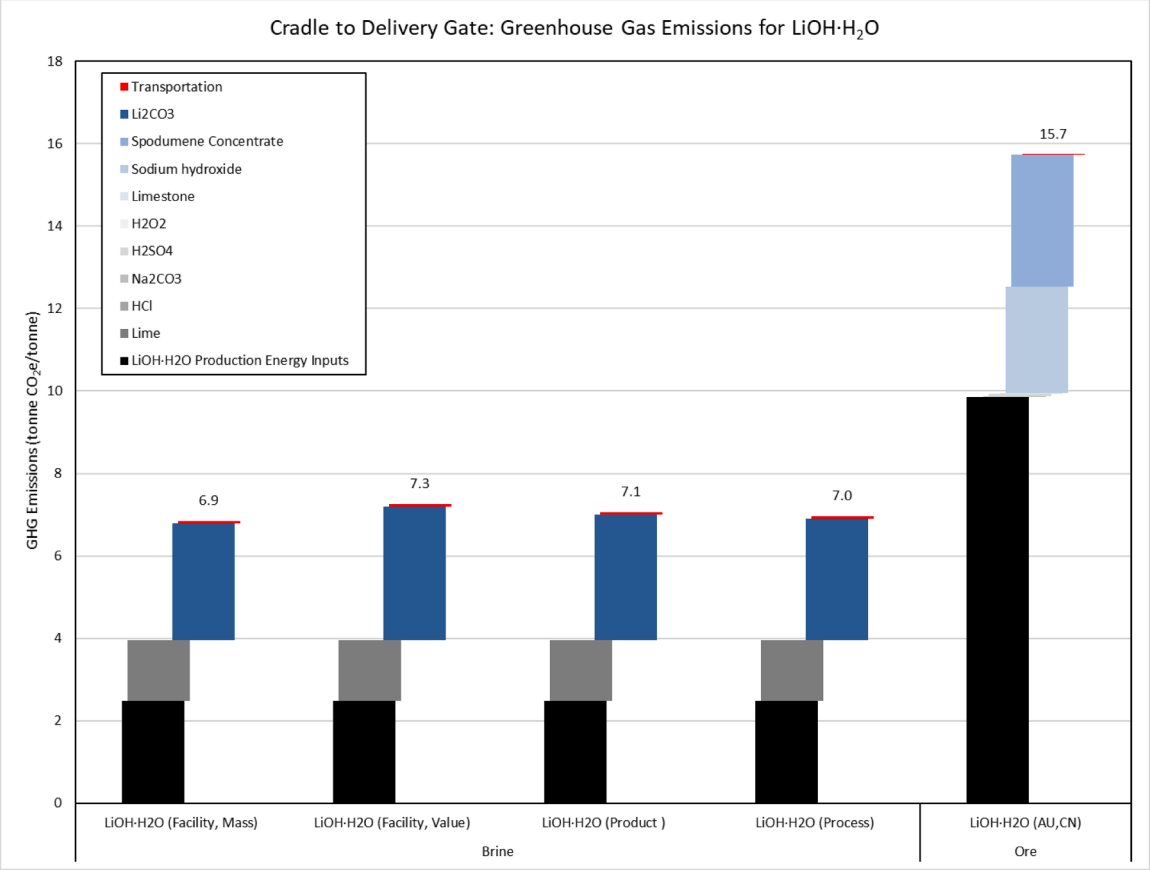


Fig. 5. Cradle-to-gate life cycle GHG results by input and process for LiOH•H_2O production based on Chilean brine and Australian spodumene ore.

4.4. *Cradle-to-gate LCA of LiOH•H_2O*

The production of LiOH•H_2O from lithium extracted from brine processed in Chile is detailed in Section 2.3. Section 3.2 covers lithium extracted from ore processed in Australia with LiOH•H_2O production in China. Energy, materials, and water requirements are given in Table 3 for brine and Table 5 for ore. Summarized LCIs for one tonne of LiOH•H_2O produced from brine in Chile or from ore sourced in western Australia and converted into LiOH•H2O in China are in the SI (Table S.8 and Table S.9, respectively).

We conducted a cradle-to-delivery-gate LCA based on these data using the GREET model. The results of the energy use are shown in Figure S.13 by type of input energy for production and transportation. The LiOH•H_2O production stage in China is very energy intensive and performs worse than the Chilean brine-based production. Heat energy for the Chinese facility is provided by coal. Figure S.14, the breakdown of energy for LiOH•H_2O production by process input, shows the energy input for material and transportation. The LiOH•H_2O process itself is very energy intensive for each case, and the embodied energy in lime and Li_2CO_3 are also intensive. For the ore-based product, both the sodium hydroxide and the spodumene concentrate contain significant embodied energy. Fig. 5 shows that the results for GHG emissions echo the results of the energy findings. Figure S.15 presents the results for water, where again the ore-based production has higher consumption than the brine-based production.

In Fig. 5, we see that the LiOH•H_2O production process is a meaningful contributor to GHG emissions in both the brine and ore pathways at ~35% of the burden for brine pathways and ~60% for the ore pathway. Again, the ore-based production is based on a facility using coal for heat, so using natural gas could reduce that burden (see SI Section 4.7.1). Additional contributors to the GHG burden for the brine pathway are lime and Li_2CO_3, while spodumene concentrate and NaOH are important contributors for the ore pathway.

The GHG results presented in Fig. 5 are in reasonable alignment with those developed by the consultancy Minviro, which conducted an LCA of LiOH•H_2O production from brine and ore (Grant et al., 2020). Their results indicate that LiOH•H_2O from Chilean brine has a CO_2 intensity of 5 tonnes of CO_2 per tonne of LiOH•H_2O. They estimate an intensity of 14.8 tonnes of CO_2 per tonne of LiOH•H_2O when the source is Australian spodumene processed in China (Grant et al., 2020). Note that their results are in presented in terms of CO_2, and they do not clearly identify whether they include GHG emissions other than CO_2, such as CH_4 and N_2O, which are included in our study.

4.5. *Cradle-to-gate LCA of NMC cathode production*

We examined the effect of lithium production routes on the life-cycle burden of lithium-ion battery cathode materials (see Stage 4 in Fig. 1), putting the lithium contribution into the context of other constituent cathode materials and production processes. We examined cathode materials NMC622 and NMC811—lithium nickel manganese cobalt oxide with a molar proportion between nickel, manganese, and cobalt of 6-2-2 and 8-1-1, respectively. These two chemistries represent a dividing line for NMC battery cathode producers: NMC622 typically utilizes the Li_2CO_3 form of lithium while NMC811 uses the LiOH•H_2O form. NMC811 cathodes hold promise for higher energy density and lower cobalt content, which matches the trend in the automotive lithium-ion battery industry. NMC622 cathode production consumes 0.381 tonnes of Li_2CO_3 per tonne of cathode (0.072 tonne of Li per tonne of cathode), and NMC811 cathode production consumes 0.431 tonnes of LiOH•H_2O per tonne of cathode (0.071 tonne of Li per tonne of cathode).

A life cycle inventory for the energy and materials inputs to both

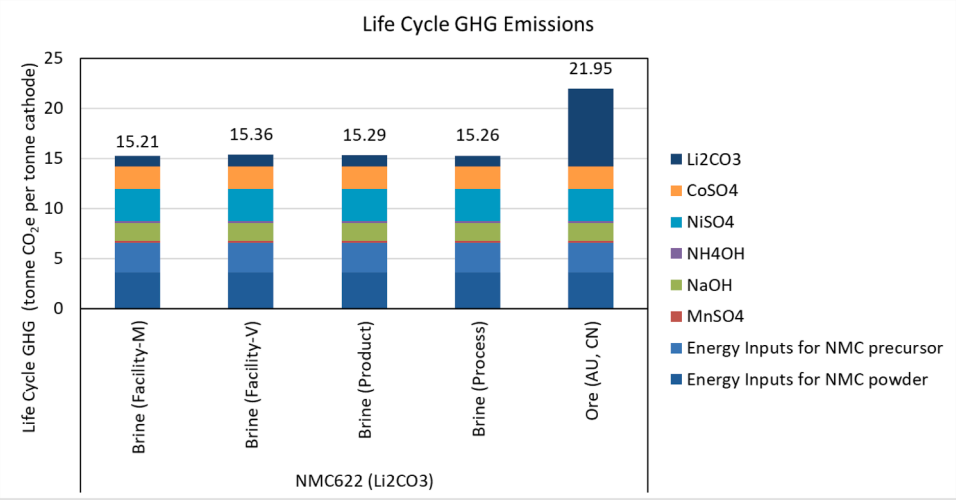


Fig. 6. Cradle-to-gate life cycle GHG results for NMC622 cathode materials with lithium from different sources.

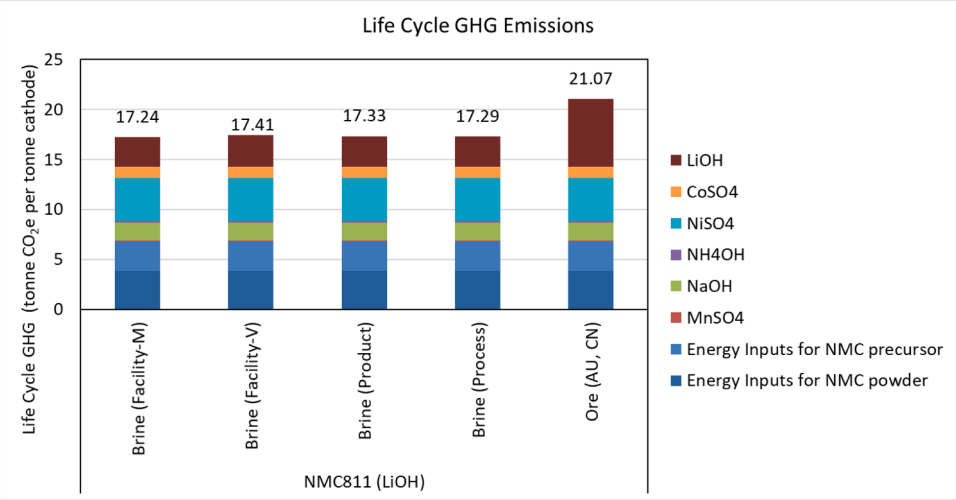


Fig. 7. Cradle-to-gate life cycle GHG results for NMC811 cathode materials with lithium from different sources.

NMC622 and NMC811 cathode material are described in Dai et al. (2021). That study utilized industrial-based data for material and energy inputs, which have been integrated into the GREET model (the basis for background data here). The per-tonne cathode inputs for NMC622 and NMC811 are presented in Table S.10. In this work, we utilize the detailed lithium and lithium chemicals production data described here in place of GREET's default lithium data. All other data related to cathode production leverage baseline GREET information (i.e., inventory data for other input materials). This allows for comparison of the effect that the different lithium sources (and allocation approaches) have on the life cycle results.

We examined the primary pathways of brine (represented by four Chilean allocation approaches) and ore (Australian ore with processing in China). Figure S.16 shows the total life cycle energy results of NMC622 cathode production on a per-tonne-cathode basis. Figure S.17 shows the same for NMC811. Results are presented by material inputs and process. Li_2CO_3 from the ore pathway is a significant contributor to the overall energy consumption in the production of NMC622, while the brine pathways are less intensive. NMC811 energy consumption suggests that the brine-sourced $LiOH•H_2O$ is less energy intensive than the ore-sourced $LiOH•H_2O$.

Results for life-cycle GHG emissions align closely with energy results for the NMC622 cathode (Fig. 6) and the NMC811 cathode (Fig. 7). Namely, we see that for the NMC622 pathway, while all other inputs are held constant, the ore-based Li_2CO_3 increases GHG emissions by ~40% compared to the brine-based. For NMC811, we see that ore-based $LiOH•H_2O$ increases GHG emissions by ~20% compared to the brine-based. It is important to recall for the ore-based products, that their emissions could be reduced if natural gas were used in place of coal for heat in the production processes.

Water consumption in the brine allocation approaches varies more, as seen in Figure S.18 for NMC622 and Figure S.19 for NMC811. The water results suggest that the production of ore-based cathodes consumes more fresh water than the production of brine-based cathodes, but it is important to note that our knowledge of water use in the ore pathway is more limited than in the brine pathway.

4.6. *Cradle-to-gate LCA of NMC automotive batteries*

Finally, we present the effect of lithium production routes on the life-cycle burden of lithium-ion batteries for vehicles with a 300-mile driving range (482 km) (Stage 5 in Fig. 1). This allows us to view the effect of lithium production on the complete battery system. We examine NMC622 and NMC811 cathode materials for use in the automotive battery. Based on Argonne's BatPaC Model, the specific energy of an NMC622 battery is 241 Wh/kg, while

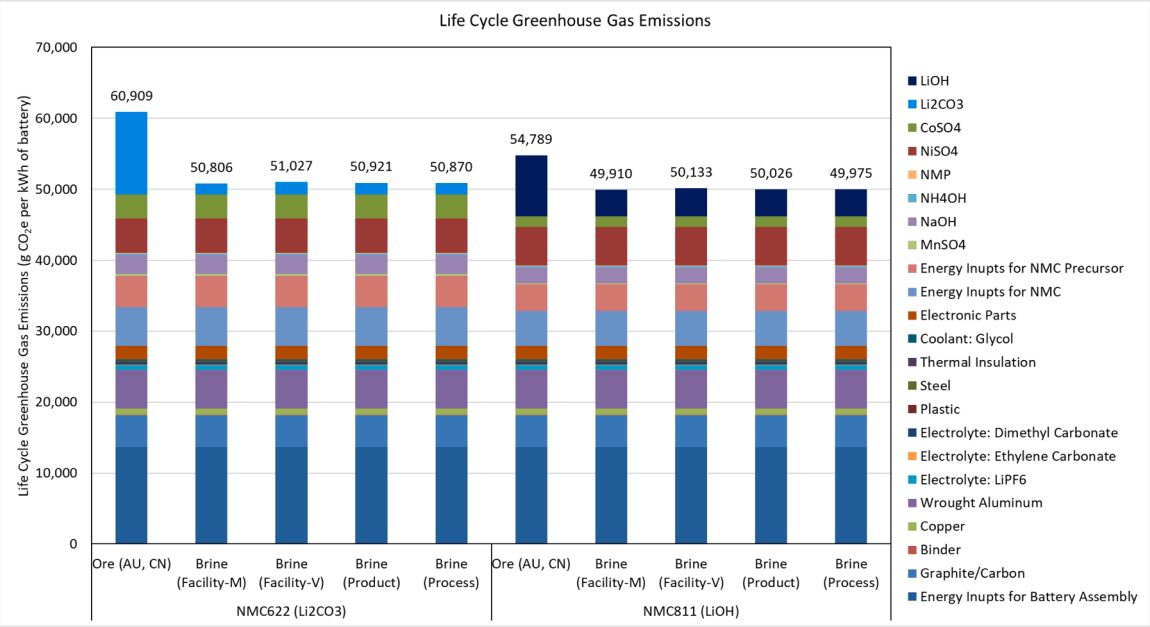


Fig. 8. Cradle-to-gate life cycle GHG results for batteries by input and process with lithium from different sources.

for an NMC811 it is 248 Wh/kg [26]. Both batteries have 84 kWh of energy capacity to achieve the 300-mile range. An NMC622 battery contains 0.57 kg Li_2CO_3 per battery kWh (0.11 kg Li per battery kWh), and an NMC811 battery contains 0.54 kg $LiOH•H_2O$ per battery kWh (0.09 kg Li per battery kWh).

Battery material inputs were modeled using Argonne BatPac Model v4.0 for both NMC622 and NMC811 (Argonne National Laboratory 2020). Winjobi et al. describe the life cycle inventory details for these batteries for GREET integration (Winjobi et al., 2020). Here, those battery bill of material and energy use details (Table S.11) are utilized along with the detailed lithium production processes described earlier to evaluate the effect that the different lithium sources (and allocation approaches) have on the life-cycle results of batteries. Again, background data are gathered from GREET's baseline settings.

In Figure S.20, we present the total life-cycle energy results of the NMC622 and NMC811 lithium-ion batteries on a per-battery-kWh basis. We examined the primary pathways of brine (represented by brine's four allocation approaches) and ore (represented by Australian ore with processing in China). We observe that the brine allocation approach has a minimal impact on total energy results, and that the brine pathway is less energy intensive than the ore pathway.

Fig. 8 shows the GHG results for battery production for the given scenarios. The GHG results closely mirror those of the energy results. Those results highlight that the effect of concentrated lithium brine allocation approach does not yield significant variance in the battery's GHG emissions, but that brine-sourced lithium yields NMC622 batteries with 20% lower emissions and NMC811 batteries with 10% lower emissions than ore-sourced lithium. Results for life cycle water consumption are presented in Figure S.21 and show meaningful variability between the scenarios, with slightly more variance between the brine allocation approaches than we see in energy consumption or GHG emissions results.

Overall, we used LCA to connect the production of lithium from brine or ore through resource concentration, production of lithium compounds (Li_2CO_3 and $LiOH•H_2O$), and production of NMC622 and NMC811 cathode powder, all the way through lithium's utilization within an electric vehicle in the form of an NMC622 or NMC811 battery (84 kWh). Our analysis shows not only that the source of the lithium matters (brine or ore) with respect to environmental effects, but also that the resource allocation approach used for brine impacts the reported environmental effect. Since this analysis was able to leverage process-level data for brine-based production, we recommend those process-level LCA results, as they are most representative of actual process energy, materials, and water usage in the production of concentrated lithium brine.

5. Summary and conclusions

This article presented LCAs of battery-grade Li_2CO_3 and $LiOH•H_2O$ produced from brine and spodumene ores. Detailed life cycle inventory data were presented for material, energy, and freshwater consumption associated with lithium acquisition; lithium concentration; production of lithium chemicals, battery cathode powders, and batteries; and associated transportation activities. Results of the LCA show that concentrated lithium brine and its associated end products can vary significantly in energy consumption, GHG emissions, and water consumption depending upon the resource allocation method used in the analysis. It is generally preferable to obtain process-level data when collecting data for an LCA, however, it may be difficult to obtain such details in practice. Regardless, it is critical to be mindful of any process emissions that may result from individual processes, which may not be apparent at the facility level, for instances CO_2 emissions associated with calcination.

We modeled brine-based lithium extracted from the Salar de Atacama in Chile and then dried in large ponds to evaporate the water, concentrate the lithium, and remove impurities. The product of that drying process is a concentrated lithium brine (~6% Li) which is transported to the coastal city of Antofagasta. There, materials and energy are added to produce Li_2CO_3. For a portion of the Li_2CO_3, additional materials and energy are added in further stages to produce $LiOH•H_2O$. The two end products, Li_2CO_3 and $LiOH•H_2O$, are then shipped worldwide to battery cathode producers where they are further processed into a variety of battery cathode materials.

We also modeled ore-based lithium extracted from spodumene ore in western Australia. The spodumene ore is first separated from the waste rock during excavation, then concentrated at a nearby location by adding materials and energy. The concentrated spodumene is transported for processing elsewhere (this analysis considers processing in China), where material and energy are combined with the spodumene concentrate through chemical processes to produce either Li_2CO_3 or

Table 6
Results of LCA for lithium concentrates and chemical products from brine and ore.

Lithium source	Stage of evaluation	GHG emissions	Energy consumption	Freshwater consumption
Brine	Lithium concentration	0.08–0.18 g CO_2e/tonne lithium concentrate	1300–2800 MJ/ tonne lithium concentrate	2.95–7.30 m^3/tonne lithium concentrate
	Production of Li_2CO_3 from lithium concentrate*	2.7 – 3.1 tonne CO_2e/ tonne Li_2CO_3	30,000–36,000 MJ/tonne Li_2CO_3	15.5 – 32.8 m^3/tonne Li_2CO_3
	Production of LiOH•H_2O from lithium concentrate	6.9 – 7.3 tonne CO_2e /tonne LiOH•H_2O	76,600–82,900 MJ/tonne LiOH•H_2O	31–50 m^3/ tonne LiOH•H_2O
Ore	Spodumene concentration	~0.42 tonne CO_2e/tonne spodumene	5500 MJ/tonne spodumene	3.4 m^3/tonne spodumene
	Production of Li_2CO_3 from spodumene*	20.4 tonne CO_2e/tonne Li_2CO_3	218,000 MJ/ tonne Li_2CO_3	77 m^3/tonne Li_2CO_3
	Production of LiOH•H_2O from spodumene	15.7 tonne CO_2e/tonne LiOH•H_2O	187,200 MJ/ tonne LiOH•H_2O	69 m^3/tonne LiOH•H_2O

* The order-of-magnitude difference in life cycle GHG emissions for Li_2CO_3 production between brine and ore is related to process differences, including energy inputs and embodied emissions materials inputs.

LiOH•H_2O. Those products are then shipped to battery cathode producers. The modeled Li_2CO_3 or LiOH•H_2O production in China utilizes industry -reported data for a facility that provides process heat from coal, thus we can expect that a transition to other heat sources could dramatically reduce total GHG emissions for those products, which would therefore reduce GHG emissions for cathodes and batteries from those products.

In both the brine and ore production pathways of this analysis, the intermediary products following the production of Li_2CO_3 and LiOH•H_2O are NMC622 and NMC811, respectively. The NMC622 and NMC811 are then incorporated with an anode, electrolyte, packaging, etc., to yield an automotive lithium-ion battery. Here a capacity of 84 kWh is assumed. The LCA in this study examined the results of the brine and ore pathways at both the endpoint (the NMC battery), and the intermediary stages associated with lithium.

LCA results for the lithium chemical products are shown in Table 6. This LCA also found that on the basis of battery cathode material, the difference in lithium source leads to as much as a 20% difference in GHGs for NMC811 cathodes and up to a 45% difference in GHGs for NMC622 cathodes. For the battery as a whole, this represents up to a 9% difference for NMC811 batteries and a 20% difference for NMC622 batteries.

This study identifies that lithium will play an increasingly important role in the life-cycle footprint of lithium-ion batteries (Fig. 8). To our knowledge, this is the first LCA of lithium compounds based on primary industry data representing two of the top global lithium production locations, so these results should be representative of current global lithium compounds production.

The study identifies key differences between brine- and ore-based lithium and their implications on the environmental impacts of lithium products. The brine-based pathway predominantly uses solar energy to concentrate Li up to 6%, while the ore-based pathway relies heavily on fossil fuels (diesel for ore mining and processing, and coal for refining 6% Li2O, which has <3% Li) to get a comparable Li concentration. This, combined with the greener electric grid in Chile versus China, leads to less GHG emissions from lithium compounds produced from brine. Further, the ore-based pathway allows LiOH•H_2O production directly from spodumene concentrate, while for the brine-based pathway LiOH•H_2O is produced by conversion of Li_2CO_3. The more streamlined process for LiOH•H_2O production from ore helps to narrow the gap between the GHG emissions from brine- and ore-sourced pathways, compared to Li_2CO_3 production. If the spodumene mining, processing, and refining processes were powered by renewable energy, LiOH•H_2O produced from ore could have significantly lower GHG emissions.

Finally, we consider opportunities to improve the life-cycle footprint of lithium chemicals. We observe that process energy inputs, especially those consumed at the lithium chemical production facilities, account for a significant portion of the GHG emissions from the production of brine- and ore-based lithium chemicals. Thus, decarbonizing the fuels and electricity consumed at these plants represents the best opportunity to reduce the GHG emissions from lithium chemical production.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Credit Author Statement

Energy, Greenhouse Gas, and Water Life Cycle Analysis of Lithium Carbonate and Lithium Hydroxide Monohydrate from Brine and Ore Resources and Their Use in Lithium Ion Battery Cathodes and Lithium Ion Batteries

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.resconrec.2021.105762.

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